Introduction to Programming

Lecture 20

VU
Virtual University

In the Last Two Lectures

- File Handling
  - Sequential Files
  - Random Access Files

Today's Lecture

- Structures
- Unions

Structure

?

Structure definition

Keyword

Structure name

```
struct Student
{
 char name [ 60 ] ;
 char address [ 100 ] ;
 double gpa ;
} ;
```

Data of the structure

Student s1 , s2 , s3 ;

Structure Name

Variable Names

Structure

```
struct address
{
  char streetAddress [ 100 ] ;
  char city [ 30 ] ;
  char country [ 30 ] ;
}
```

Structure

```
struct Student
{
  char name [ 60 ] ;
  address add ;
  double gpa ;
} s1 , s2 , s3 ;
```

Example 1

```
struct Card
{
  char *suit ;
  char *value ;
} ;
```

Structures

Variables of type structure

Pointers to Structure

Array of Structure

Structures

```
Student s [ 100 ] ;
Student *sPtr ;
```

Structures

```
Student stdnt1 , stdnt2 ;
stdnt1 + stdnt2 ; //Wrong
```

Structures

Student s1 , s2 ;

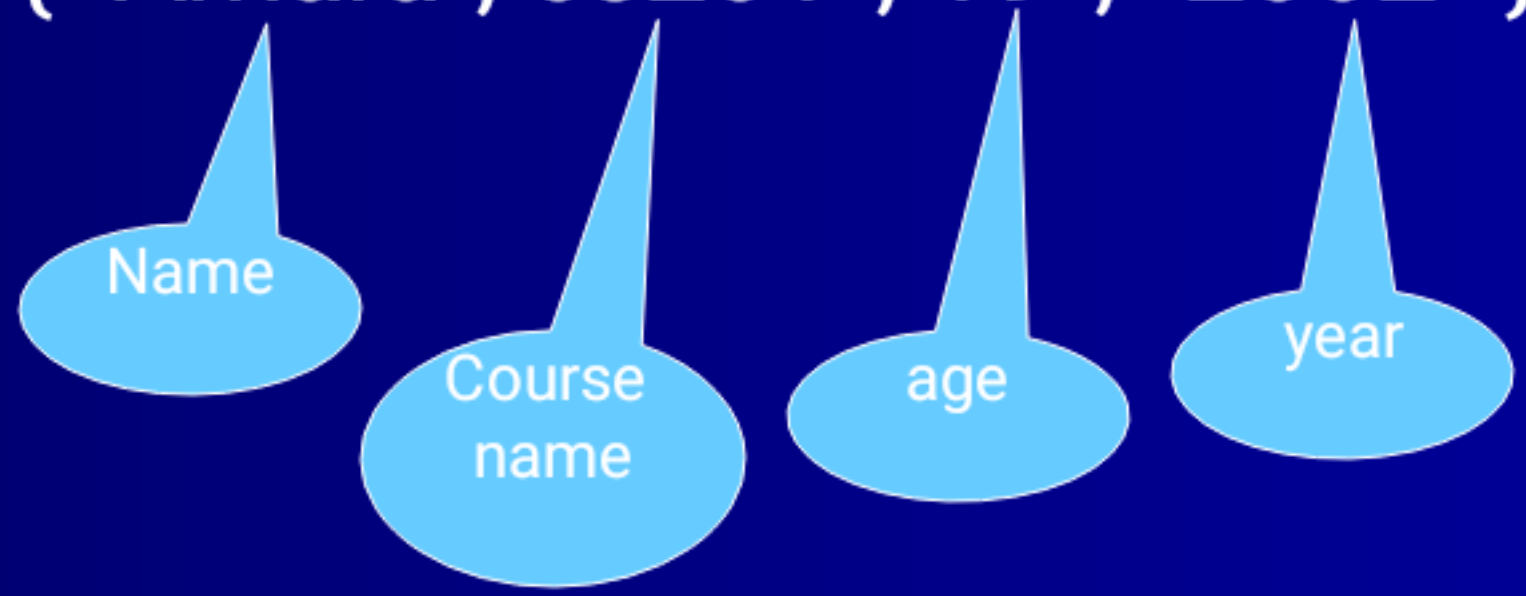
s1 = s2 ;

Example 2

```
struct Student
{
  char name [ 64 ] ;
  char course [ 128 ] ;
  int age ;
  int year ;
} ;
```

Initializing Structures

Student s1 = { “Amara”,“cs201”,“19”, “2002” } ;



Initializing Structures

s1.name ;

s1.course ;

s1.age ;

s1.year ;

Initializing Structures

```
s1.age = 20 ;
s1.name = “Abbas Ali ” ; //Wrong
strcpy ( s1.name , “Abbas Ali ” ) ;
```

Accessing structure members

```
cout << s1.name ;
cout << s1.course ;
cout << s1.age ;
cout << s1.year ;
```

Assignment of structures

```
Student s1 , s2 ;
s2 = s1 ;
```

Passing Structures to Functions

Passing Structures to Functions

Call by value

Call by Reference X

```
int Square ( int x ) ;
```

```
x = Square ( 10 ) ;
```

Structures

- Simple Variable of type Structure
- Pointer to Structure
- Arrays of Structures
- Function that returns a Structure
- Pass the Structure to functions

Pointers to Structure

Pointers to Structure

```
Student *sPtr , s1 ;
sPtr = &s1 ;
```

Pointers to Structure

sPtr.name ;

Wrong

Pointers to Structure

*sPtr.name ;

Pointers to Structure

*(sPtr).name ;

Pointers to Structure

*sPtr ->name ;

Same as

s1.name

Arrays of Structures

Arrays of Structure

```
Student s [ 100 ] ;
s [ 0 ].name ;
s [ 1 ].name ;
s [ 2 ].name ;
.
.
.
s [ 99 ].name ;
```

sizeof (s1) ;

Example 3

```
struct Student
{
 char firstName [ 30 ] ;
 char lastName [ 30 ] ;
 char course [ 15 ] ;
 char rollNo [ 10 ] ;
 int age ;
 float gpa ;
} ;
```

Example 3

```
Student s [ 10 ] ;
for ( int i = 0 ; i < 10 ; i ++ )
{
 cout << “Please enter the student's last name : " ;
    cin >> s [ i ].lastName ;
    cout << “Please enter the student's first name : " ;
    cin >> s [ i ].firstName ;
    cout << " Please enter the student's course : " ;
    cin >> s [ i ].course ;
    cout << " Please enter the student's Roll No. : " ;
    cin >> s [ i ].rollNo ;
    cout << " Please enter the student's grade : " ;
    cin >> s [ i ].grade ;
    cout << " Please enter the student's age : " ;
    cin >> s [ i ].age ;
    cout << " Please enter the student's GPA : " ;
    cin >> s [ i ].gpa ;
}
```

Example 4

Student getData ( ) ;

Return Type

Function Name

Example 4

```
ifstream inFile ( “myText.txt” ) ;
void main ( )
{
  Student s1 ;
 s1 = getData ( ) ;
 inFile.close ( ) ;
}
```

Example 4

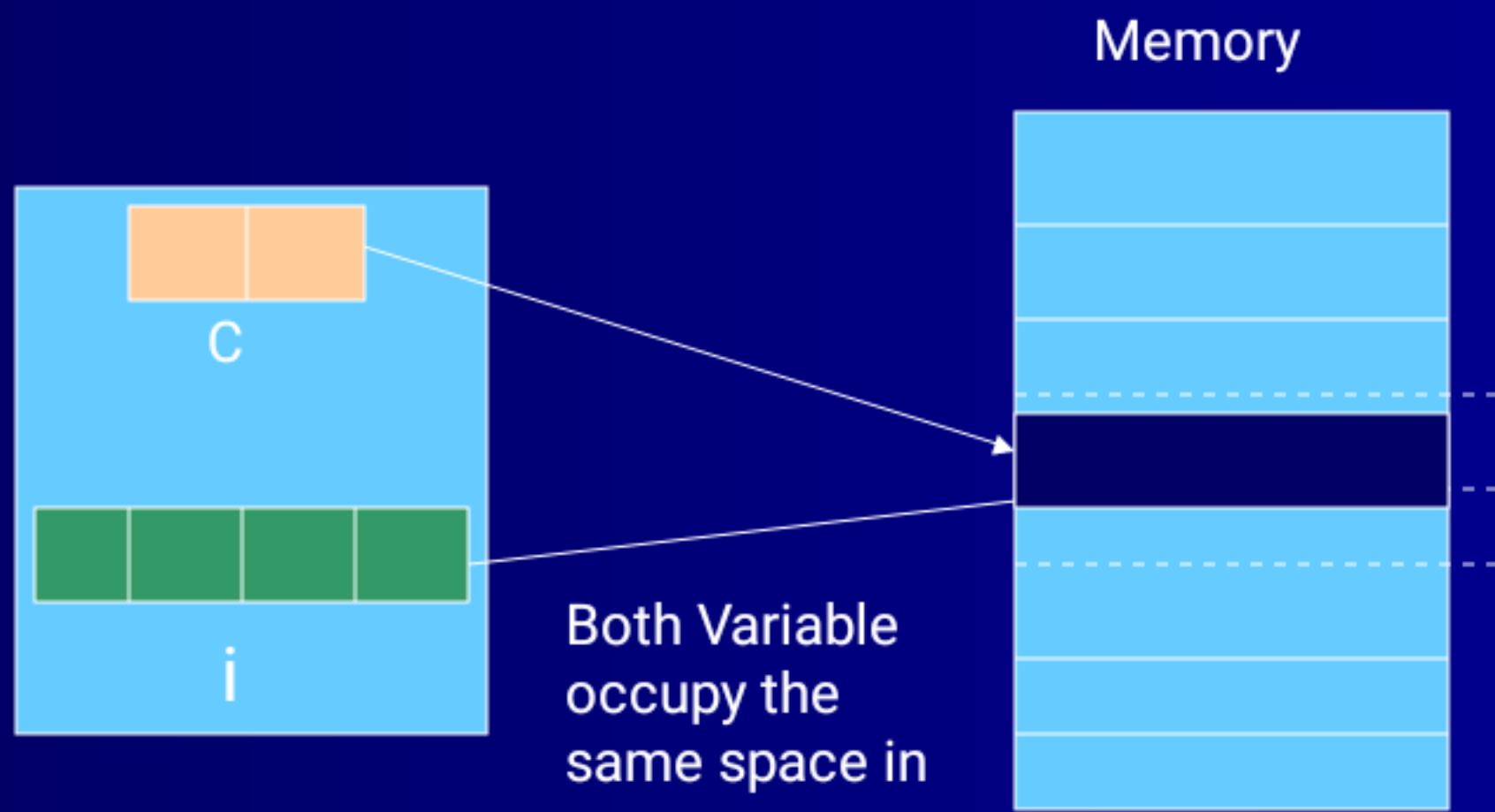
```
Student getData ( )
{
   Student tempStudent ;
 inFile >> tempStudent.firstName ;
 inFile >> tempStudent.lastName ;
 inFile >> tempStudent.course ;
 inFile >> tempStudent.rollNo ;
 inFile >> tempStudent.age ;
 inFile >> tempStudent.gpa ;
 return tempStudent ;
}
```

Union

Union

```
union intOrChar
{
  int i ;
  char c ;
} ;
```

Union

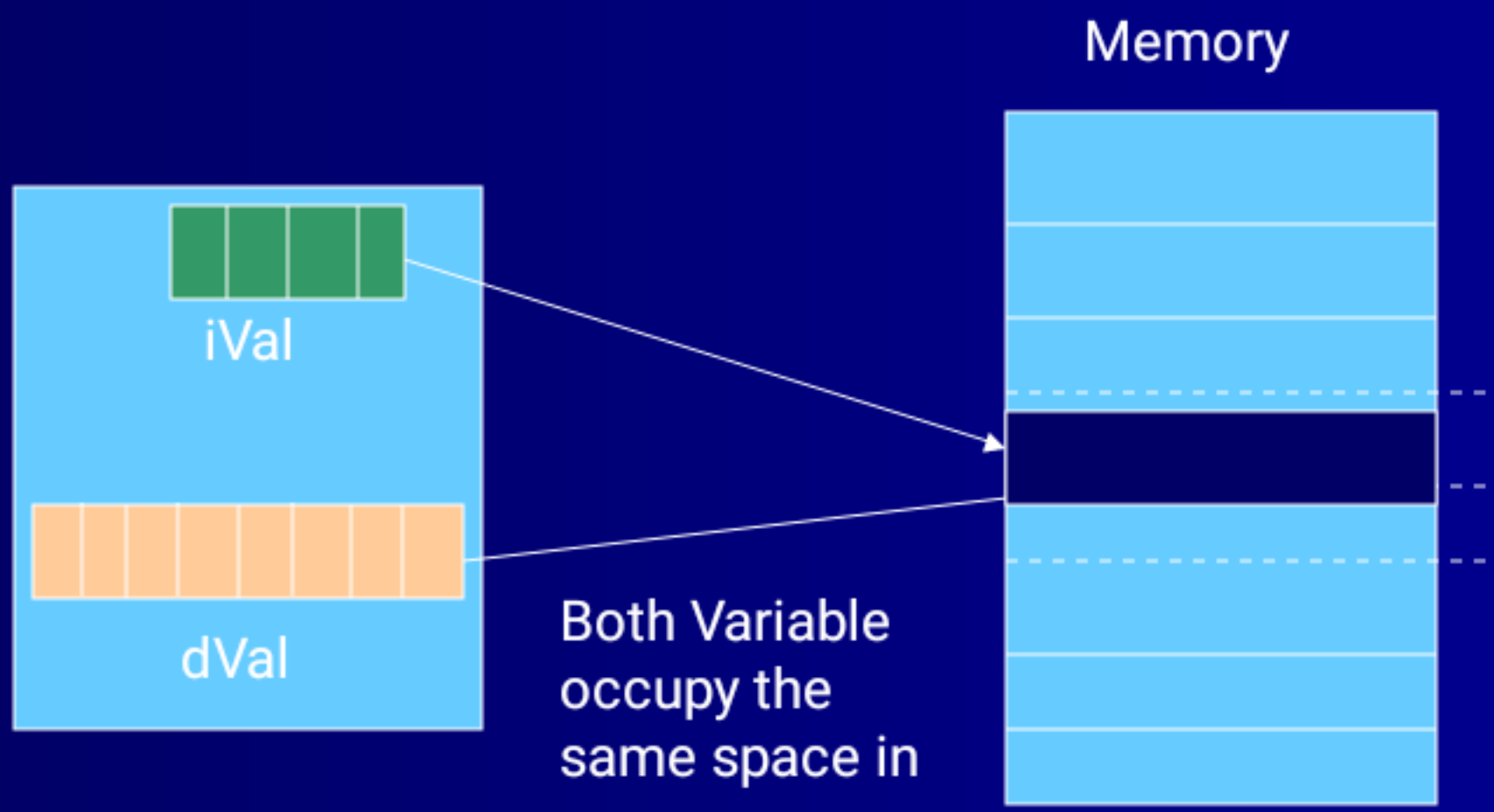


Both Variable occupy the same space in memory

Union

```
union intOrDouble
{
 int ival ;
 double dval ;
} ;
```

Union


Memory
iVal
dVal
Both Variable occupy the same space in memory

Union

```
intOrDouble u1 ;
u1.ival = 10 ;
cout << u1.ival ;
cout << u1.dval ;  incorrect value
```

Union

u1.dval = 100.0 ;

cout << u1.ival ; incorrect value

Example 5

```
char c ;
int x ;
x = 'a' ;
cout << x ;
x = x * 256
cout << x ;
  :
```

Example 5

- x = x +'b' ;
- cout << x ;

Example 5

```
Union iandc
{
 char c [ 4 ] ;
 int x ;
} ;
```

Today we studied

- Structures
- Unions